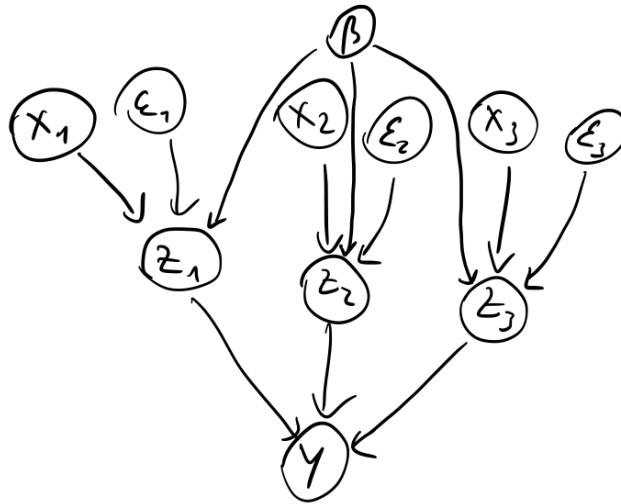


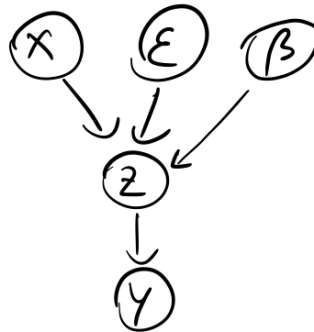
Homework 4

Task 1

I have drawn the graph once in an aggregated manner and once disaggregated.



or



Task 2

Look at code 'Task2.R'

Task 3

I wrote code for this task, see `Augmenting_z.R`. The main idea is that once we find the maximum $z_{j,i}$ with the indicator function, we resample the z 's based on

the observed y_i and introduce truncation of the $z_{k,i} \neq z_{j,i}$.

In the Multinomial Probit (MNProbit) model, we need the full conditional distributions for: 1. The latent variables $z_{j,i}$, given the observed data y_i and parameters β . 2. The parameter vector β , given the latent variables $z_{j,i}$ and predictors $x_{j,i}$.

1. Full Conditional Distribution of $z_{j,i}$

The latent variables $z_{j,i}$ are modeled as:

$$z_{j,i} = x_{j,i}^\top \beta + \epsilon_{j,i}, \quad \epsilon_{j,i} \sim N(0, 1)$$

Key Idea

- We observe y_i , which indicates the index of the maximum $z_{j,i}$.
- Therefore, $z_{j,i}$ is **truncated** based on y_i :
 - If $y_i = j$, then $z_{j,i} > z_{k,i}$ for all $k \neq j$ (meaning $z_{j,i}$ is unrestricted above).
 - If $y_i \neq j$, then $z_{j,i} \leq z_{y_i,i}$ (this means $z_{j,i}$ is restricted to lie below the maximum).

Conditional Distribution

The full conditional distribution for $z_{j,i}$ is:

$z_{j,i} | y_i, x_{j,i}, \beta \sim \text{Truncated Normal}(x_{j,i}^\top \beta, 1, L, U)$ The mean of the untruncated normal distribution is

$$x_{j,i}^\top \beta.$$

The standard deviation is 1 and the truncation bounds L and U are determined as:

- For the chosen alternative $j = y_i$:

$$L = \max_{k \neq j} z_{k,i}, \quad U = \infty$$

- For the non-chosen alternatives

$$j \neq y_i$$

:

$$L = -\infty, \quad U = z_{y_i,i}$$

Task 4 and 5

I used the provided R file `rbprobitGibbs` as a basis for creating a Gibbs sampler. I had some trouble creating a sampler that runs without errors.

The last part of the file simulates data for the binary probit model. The `n` corresponds to the number of observations, the `p` determines the number of predictors (dimension of β). Then I randomly set the true parameters of β and try to recover them using the Gibbs Sampler. As a result, the Gibbs Sampler recovered parameters that were quite close to the true ones. Small deviations might be explained by the specification of priors.

Parameter	True Beta	Posterior Mean Beta
<code>beta_1</code>	-0.424845	-0.4821123
<code>beta_2</code>	0.5766103	0.6757679
<code>beta_3</code>	-0.1820462	-0.1705051
<code>beta_4</code>	0.7660348	0.7516217
<code>beta_5</code>	0.8809346	1.098932

Task 6

As I use markdown and just compile to pdf, the Latex Notation causes some issues sometimes. Therefore; I formatted in latex and just copy and pasted screenshots from that in here:

1 Introduction

Augmentation of z

Conditioned on the observed outcome $y_i = j$, the constraints on z are:

$$z_{j,i} > z_{k,i} \quad \forall k \neq j.$$

This implies that z lies in a truncated region of the multivariate normal distribution:

$$z_i \sim N(X_i\beta, \Sigma), \quad \text{truncated to } \{z_{j,i} > z_{k,i}, \forall k \neq j\}.$$

Conditional Distribution

The full conditional distribution of z_i is:

$$z_i \mid y_i, X_i, \beta \sim \text{TruncatedMultivariateNormal}(X_i\beta, \Sigma),$$

where the truncation region is defined by the inequalities:

$$z_{y_i,i} > z_{k,i}, \quad \forall k \neq y_i.$$

Steps for Sampling z

To augment z , follow these steps:

1. Compute the mean $\mu = X_i\beta$ and covariance Σ for the multivariate normal distribution.
2. Sample z_i from the multivariate normal distribution truncated to the region defined by $z_{y_i,i} > z_{k,i}$ for all $k \neq y_i$.

Efficient Sampling with Gibbs Steps

Since ϵ is correlated ($\Sigma \neq I$), the multivariate normality must account for the covariance structure. To sample efficiently:

1. Use the conditional properties of the multivariate normal:

$$z_{j,i} \mid z_{-j,i}, X_i, \beta \sim N(\tilde{\mu}_{j,i}, \tilde{\sigma}_{j,i}^2),$$

where:

$$\begin{aligned}\tilde{\mu}_{j,i} &= X_{j,i}^\top \beta + \Sigma_{j,-j} \Sigma_{-j,-j}^{-1} (z_{-j,i} - X_{-j,i} \beta), \\ \tilde{\sigma}_{j,i}^2 &= \Sigma_{j,j} - \Sigma_{j,-j} \Sigma_{-j,-j}^{-1} \Sigma_{-j,j}.\end{aligned}$$

2. Sample each $z_{j,i}$ iteratively, ensuring that the truncation constraints $z_{y_i,i} > z_{k,i}$ are satisfied.

Final Result

The augmented z_i is drawn from:

$$z_i \mid y_i, X_i, \beta \sim \text{TruncatedMultivariateNormal}(X_i \beta, \Sigma),$$

with truncation defined by the observed y_i .